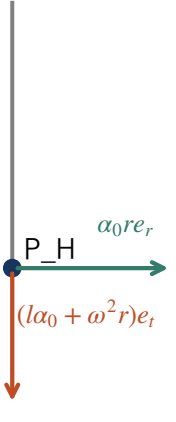
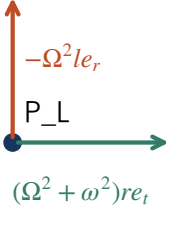
最高位置：欧拉项沿摆臂

$$a_H^2 = (l\alpha_0 + \omega^2 r)^2 + (\alpha_0 r)^2$$



最低位置前视图最左点：科里奥利项为零

$$\rho_L = -re_t, \quad v_{rel} \parallel e_k$$